**CIDGOH**
Blusson Hall, Simon Fraser University, Burnaby
www.cidgoh.ca www.imicroseq.ca

# Needs Assessment Interview Questions

**Project Name**: iMicroSeq Data Standards Development
**Interviewee Name**:
**Affiliation / Role**:
**Date & Time**: 2025

## Overview

### Summary of the Project/Pilot aims and context

The Centre for Infectious Disease Genomics and One Health (CIDGOH) has been engaged to develop a data specification for the iMicroSeq Project, a collaborative effort led by Dr Emma Griffiths (CIDGOH) and Dr Fiona Brinkman (Brinkman Lab).

The goal of this project is to assess the current (contextual) data ecosystem supporting the iMicroSeq project, specifically by identifying what data is being collected, how it is managed and used, and any downstream processing or sharing.

### Purpose of the interview

To identify current data practises, data size, type and variety, as well as the processes involved to inform the needs assessment component of developing a data standard. We aim to enhance collaboration, improve data quality, and support informed decision-making in iMicroSeq.

### Consent (if applicable)

Notes to be taken, no recording.

## Stakeholder context

| | Details |
|---|---|
| *Job Title* | Researcher |
| *Organisation type* | (e.g. Academia Government, Policy, Agriculture etc) |
| *Role in data lifecycle* | (e.g. sample/data collection, curation, analysis, use, dissemination etc) |
| *Types of data currently collected* | (fastq, xlsx etc) |

# Needs Assessment

The following questions ….

## Scope

1. What are the questions your project/initiative is trying to answer?
   *Research, surveillance? Pathogen/target?*
   *Assay - amplicon, metagenomics, marker-based, culture & WGS?*

2. Who are your partners?
   *Informs data flow, sharing, harmonization needs*

## Sampling Strategy

3. What kinds of samples are you collecting?
   *Hosts (anatomical materials/parts/body products)*
   *Environments (materials/sites)*
   *Collection devices/methods*

4. Are you using any previously collected samples?
   *Purpose of sampling/sample bias, data reuse*

5. Do you have a sample plan? Is the sample plan, experimental design, or any other material available online?

### Sample Collection

6. What contextual data associated with the samples is being collected/considered?
   a. Hosts - age/gender/exposures
   b. Physico-chemical attributes? Weather?
   c. Land use, water table and/or tidal data, zone info (benthic/pelagic)?
   d. Presampling activities
   e. Experimental interventions?
   f. Range of geo-locs and other geo-spatial data
   g. Sampling frequency and duration

## Specimen processing and Storage

7. Are the samples subjected to any special processing before sequencing libraries are prepared?
   *Filtering, enrichment, culturing, etc*

8. If samples are cultured, what kinds of media and conditions are being used?

9. How are samples stored, and for how long
   *Cold chain, comparability, audit*

## Library Preparation

10. What are your library preparation processes?
    *Extraction, enrichment*

11. How are libraries being generated?

12. What are your controls?
    *endogenous controls, replicates, synthetic constructs/data*

## Sequencing and Bioinformatics

13. What sequencing instruments are being used?

14. What bioinformatics processing is being performed
    a. What type of processing is performed?
       *Tailored scripts? Community led pipelines/workflows?*
    b. Tools
       i. Raw sequence processing/dehosting
       ii. Consensus
       iii. Callers
    c. Databases (reference or otherwise)
    d. QC metrics

## Associated Data Types

15. Are there other data types or characterizations available for the samples/sequences?
    a. Physico-chemical properties:
       *Temp, pH, oxygen, salinity, nitrogen*
    b. Biological properties:
       *Enzymatic, staining, marker identification, colony counts, AMR, typing (serotype), matter composition, mollusc shell length, therapeutic history, etc*

## Data Management

16. How is the contextual data being captured?
    *Spreadsheets, LIMS, RedCap, other?*

17. Are any data standards being used to structure the data? Yes/No
    a. If no, would your partners be interested in using international standards?

18. Where is the data being stored?Locally, public repositories, network databases?

19. What is the data flow (where does it need to go, how does it get here)

20. Among your network(s), are there any data use limitations/restrictions? Due to:
    a. Limitations/biases of the data
    b. Data governance

21. During data flow, will the data need to be transformed?
    *Curation and automation needs*

22. Is there a plan to share the data publicly? If no, why? What are the perceived challenges?
    *Governance/privacy/equity/policies/technical issues*

## Future Work (anticipating future needs)

23. What data types would you want to collect/foresee integrating in the future?

24. What are your main contextual data challenges?

25. What would be on your contextual data wishlist?

# Summary

Summary to be included here

| **Data ecosystem** | **Details** |
|---|---|
| *Types of data currently collected* | |
| *Current data workflow* | |
| *Systems, standards and/or tools currently used* | |
| *Key outputs/reports that data supports* | |

| **Challenges** | **Details** |
|---|---|
| *Pain points* | |

| | |
|---|---|
| *Missing data fields* | |
| *Integration challenges* | |
| *Data governance and permissions* | |